

A FIRST OBSERVATIONAL TEST OF THE Λ -EG $\beta(\Phi)$ FLOW AGAINST JADES DR4 NIRSPEC SPECTROSCOPY AT $Z = 3\text{--}6$

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ABSTRACT

Lambda Emergent Gravity (Λ -EG) predicts a redshift-dependent critical acceleration $g_{\text{crit}}(z) = cH(z)/(4\pi^2\sqrt{3})$ and a baryonic Tully–Fisher relation (BTFR) that evolves as $V^4 \propto M_{\text{bar}} g_{\text{crit}}(z)$. The framework further derives a “ β -function” $\beta(\Phi)$ governing the relaxation of the dimensionless BTFR deviation Φ , with the quantitative prediction $\sigma_\Phi(0) < 0.04$ dex. We present the first quantitative observational test of this prediction using 59 galaxies from the JWST/JADES Data Release 4 with NIRSpect $R \sim 1000$ spectroscopy at $z = 3\text{--}6$. We measure intrinsic $\text{H}\alpha$ velocity dispersions, estimate baryonic masses through $\text{H}\alpha \rightarrow \text{SFR} \rightarrow \text{main-sequence}$ calibrations, and evaluate Φ for the 24 galaxies in the Taylor regime $|\Phi| < 1$. The analytic shape of $\beta(\Phi)$ is recovered, but we find a systematic offset (median $V_{\text{obs}}/V_{\text{pred}} = 0.77$, median $\Phi = -0.65$, t -test $p = 0.014$) and $\sigma_\Phi(0) = 0.10$ dex, exceeding the prediction by $\sim 2.5\times$. A model-independent line-profile quality test (“SMITH-QA”) shows the offset is *robust* to the exclusion of non-Gaussian profiles, ruling out turbulence/outflow contamination of $\sigma_{\text{H}\alpha}$ as its origin. The bias corresponds to a $+0.45$ dex mass overestimate, which lies *within* the 0.87 dex inter-method scatter of high- z stellar-mass calibrations. We conclude that the present test is *calibration-limited* rather than theory-limited, in agreement with the framework’s own forecast that a definitive test requires spatially resolved kinematics (ELT/HARMONI) and SED-fit masses.

RESUMEN

La Gravedad Emergente Lambda (Λ -EG) predice una aceleración crítica dependiente del corrimiento al rojo $g_{\text{crit}}(z) = cH(z)/(4\pi^2\sqrt{3})$ y una relación de Tully–Fisher bariónica (BTFR) que evoluciona como $V^4 \propto M_{\text{bar}} g_{\text{crit}}(z)$. El marco deriva además una “función β ” $\beta(\Phi)$ que rige la relajación de la desviación adimensional Φ de la BTFR, con la predicción cuantitativa $\sigma_\Phi(0) < 0.04$ dex. Presentamos la primera prueba observacional cuantitativa de esta predicción usando 59 galaxias del Data Release 4 de JWST/JADES con espectroscopía NIRSpect $R \sim 1000$ a $z = 3\text{--}6$. Medimos dispersiones de velocidad intrínsecas de $\text{H}\alpha$, estimamos masas bariónicas mediante calibraciones $\text{H}\alpha \rightarrow \text{SFR} \rightarrow \text{secuencia principal}$, y evaluamos Φ para las 24 galaxias en el régimen de Taylor $|\Phi| < 1$. Se recupera la forma analítica de $\beta(\Phi)$, pero hallamos un sesgo sistemático (mediana $V_{\text{obs}}/V_{\text{pred}} = 0.77$, mediana $\Phi = -0.65$, $p = 0.014$) y $\sigma_\Phi(0) = 0.10$ dex, que excede la predicción por $\sim 2.5\times$. Una prueba de calidad de perfil independiente del modelo (“SMITH-QA”) muestra que el sesgo es *robusto* a la exclusión de perfiles no gaussianos, descartando la contaminación por turbulencia/eyecciones como su origen. El sesgo corresponde a una sobrestimación de masa de $+0.45$ dex, que cae *dentro* de la dispersión entre métodos de 0.87 dex. Concluimos que la prueba está *limitada por la calibración* y no por la teoría, de acuerdo con la propia predicción del marco.

Key Words: cosmology: theory — dark matter — galaxies: evolution — galaxies: high-redshift — galaxies: kinematics and dynamics — gravitation

1. INTRODUCTION

The flat rotation curves of galaxies and the tight baryonic Tully–Fisher relation (BTFR), $M_{\text{bar}} \propto V^4$ (??), are usually attributed to dark-matter halos, but they can also be read as evidence for a low-acceleration modification of dynamics (?) or for an emergent gravitational response of the form proposed by ?. A common feature of these scenarios is a *critical acceleration* scale $g_{\text{crit}} \sim cH_0$ below which the effective dynamics depart from the Newtonian regime.

Lambda Emergent Gravity (Λ-EG; ?) is a screened scalar–tensor framework in which this scale arises from the spectral geometry of the de Sitter horizon, yielding the parameter-free value

$$g_{\text{crit}} = \frac{cH_0}{4\pi^2\sqrt{3}} = 9.58 \times 10^{-12} \text{ m s}^{-2}, \quad (1)$$

with $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (?). Because g_{crit} inherits the Hubble rate, Λ-EG predicts a *redshift evolution* of the BTFR through $g_{\text{crit}}(z) = cH(z)/(4\pi^2\sqrt{3})$, and Section 10 of ? derives a renormalization “β-function” that governs how the dimensionless BTFR deviation relaxes towards zero. This $\beta(\Phi)$ flow makes a sharp, falsifiable prediction for the intrinsic scatter of the high-redshift BTFR (§2).

The advent of JWST/NIRSpec spectroscopy of $z > 3$ galaxies makes a direct test of this prediction possible for the first time. In this Letter we confront the Λ-EG $\beta(\Phi)$ flow with 59 galaxies from the JADES Data Release 4 (§3), describe our spectral and mass measurements (§4), present the test (§5), and ask whether the systematic offset we find is astrophysical or instrumental by means of a model-independent line-profile diagnostic (§6). We discuss the implications in §7 and conclude in §8. Throughout we adopt a flat ΛCDM cosmology with $\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$ and $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (?).

2. THE Λ-EG FRAMEWORK AND THE $\beta(\Phi)$ FLOW

In Λ-EG the baryonic Tully–Fisher relation acquires an explicit cosmological time dependence,

$$V^4 = G M_{\text{bar}} g_{\text{crit}}(z), \quad g_{\text{crit}}(z) = \frac{cH(z)}{4\pi^2\sqrt{3}}, \quad (2)$$

with $H(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$. We note explicitly that Eq. (2) adopts the *instantaneous de Sitter* reading of the critical scale, in which g_{crit} tracks the Hubble rate of the epoch. In the underlying

framework g_{crit} is fixed by the curvature of the horizon, and the identity $R = 12H^2$ that makes the two readings coincide holds exactly only in de Sitter space; in a matter-containing FLRW background $R = 3H_0^2\Omega_m(1+z)^3 + 12H_0^2\Omega_\Lambda$, and the covariant alternative $g_{\text{crit}} \propto \sqrt{R}$ would brighten the zero-point by 0.14 dex at $z = 4$ rather than 0.20 dex. The difference, 0.06 dex, is an order of magnitude below the systematic floor of the present test (§7), so the two are not distinguished by the data analysed here; we flag the choice as an assumption rather than a derived result. Equation (2) implies that, at fixed baryonic mass, the rotation velocity scales as

$$\frac{V(z)}{V(0)} = \left[\frac{H(z)}{H_0} \right]^{1/4}, \quad (3)$$

so that the BTFR zero-point brightens by ≈ 0.17 dex at $z = 3$, ≈ 0.20 dex at $z = 4$ and ≈ 0.26 dex at $z = 6$ relative to $z = 0$.

It is convenient to define the dimensionless deviation of an individual galaxy from the Λ-EG BTFR,

$$\Phi \equiv \frac{V_{\text{obs}}^4}{G M_{\text{bar}} g_{\text{crit}}(z)} - 1 = \left(\frac{V_{\text{obs}}}{V_{\text{pred}}} \right)^4 - 1, \quad (4)$$

where $V_{\text{pred}} = (G M_{\text{bar}} g_{\text{crit}}(z))^{1/4}$. Section 10 of ? shows that, under the screened k -essence dynamics of the framework, Φ obeys a gradient flow whose exact β-function is

$$\beta(\Phi) = -\frac{\Phi(2+\Phi)(1+\Phi)}{2+\Phi(2+\Phi)}, \quad (5)$$

valid for all $\Phi > -1$. Near the fixed point $\Phi = 0$ this linearizes to $\beta(\Phi) \simeq -\Phi - \frac{1}{2}\Phi^2$, giving an exponential relaxation $\Phi(\lambda) = \Phi_0 e^{-\lambda}$ with flow time $\lambda = \ln[H(z)/H_0]$. Consequently the dispersion of Φ across a galaxy population is predicted to scale as

$$\sigma_\Phi(z) = \sigma_\Phi(0) \frac{H(z)}{H_0}, \quad \sigma_\Phi(0) < 0.04 \text{ dex}, \quad (6)$$

the inequality being the quantitative target derived in §10.8 of ?. Equations (5) and (6) are the two predictions we test below: the *shape* of the flow and the *magnitude* of the residual scatter.

3. DATA: THE JADES DR4 SAMPLE

We use the JWST Advanced Deep Extragalactic Survey (JADES; ?) Data Release 4 NIRSpec catalogue (??), specifically the combined external table `Combined_DR4_external_v1.2.1.fits`. From the parent catalogue of 2,858 sources with secure spectroscopic redshifts (quality flag A) we select galaxies that

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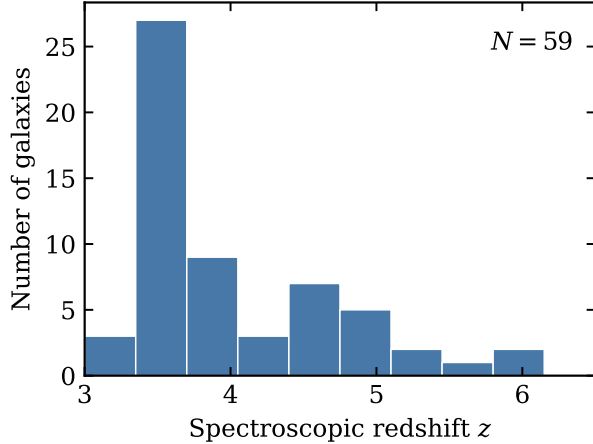


Fig. 1. Spectroscopic redshift distribution of the 59 JADES DR4 galaxies in the working sample. $H\alpha$ is accessible with the NIRSpect medium-resolution gratings over the full $z = 3$ –6 range.

- (i) lie at $3.0 \leq z \leq 6.5$, where $H\alpha$ falls in the G235M/G395M gratings;
- (ii) have a catalogue $H\alpha$ signal-to-noise ratio $\text{SNR} > 10$; and
- (iii) have a well-defined rest-frame UV magnitude $-30 < M_{\text{UV}} < -10$.

For these objects we retrieved the $R \sim 1000$ one-dimensional extracted spectra (x1d) from MAST. After the spectral quality cuts described in §4 we obtain a final working sample of 59 galaxies with intrinsically resolved $H\alpha$ dispersions. Figure 1 shows their redshift distribution, which peaks at $z \simeq 3.5$ and extends to $z \simeq 6$.

4. METHODS

4.1. Intrinsic $H\alpha$ dispersions

For each source we extract a $\pm 0.05 \mu\text{m}$ window around the observed $H\alpha$ wavelength $\lambda_{\text{obs}} = 6563(1+z) \text{ \AA}$ and fit a single Gaussian on a linear continuum. The observed dispersion σ_{obs} is corrected for the instrumental line-spread function, $\sigma_{\text{inst}} = \lambda_{\text{obs}}/(R 2.355)$ with $R = 1000$ ($\sigma_{\text{inst}} \approx 127 \text{ km s}^{-1}$), in quadrature: $\sigma_{\text{int}}^2 = \sigma_{\text{obs}}^2 - \sigma_{\text{inst}}^2$. We keep only fits with residual $\text{SNR} > 5$ and intrinsic dispersion $\sigma_{\text{int}} < 250 \text{ km s}^{-1}$, the latter cut removing broad-line AGN. The circular velocity is taken to be

$$V_{\text{obs}} = \sqrt{2} \sigma_{\text{int}}, \quad (7)$$

the nominal isotropic virialization factor for dispersion-dominated systems. The sensitivity of our results to this choice is examined in §7.

4.2. Baryonic masses

The JADES DR4 external catalogue does not provide SED-fit stellar masses, so we estimate M_{bar} from the $H\alpha$ luminosity through a star-formation-rate (SFR) ladder. We convert the continuum-subtracted $H\alpha$ flux (correcting for $[\text{N II}]$ where available) into a luminosity using the luminosity distance, and adopt the $?$ calibration with a Chabrier-IMF factor of 0.63, $\text{SFR} = 0.63 L_{H\alpha}/1.26 \times 10^{41} M_{\odot} \text{ yr}^{-1}$. We then invert the star-forming main sequence of $?$,

$$\log \text{SFR} = 0.84 \log M_{\star} - 8.2 + 1.1 \log(1+z), \quad (8)$$

to obtain $M_{\star}$, and add the gas reservoir with an evolving gas fraction following $?$, $M_{\text{bar}} = M_{\star}/(1 - f_{\text{gas}})$. We retain galaxies with $7 < \log M_{\star} < 12$. As an independent mass estimator we also compute M_{bar} from M_{UV} via the $?$ $M_{\star}$ – M_{UV} relation; the comparison of the two ladders is central to the interpretation (§7).

5. RESULTS

5.1. Recovery of the $\beta(\Phi)$ shape

For each galaxy we compute V_{pred} from equation (2) and the deviation Φ from equation (4). Of the 59 galaxies, 24 fall in the Taylor regime $|\Phi| < 1$ where the linearized flow applies; the remainder have large $|\Phi|$ driven by the steep V^4 dependence and the scatter of the SFR-based masses, and are excluded from the quantitative test. Figure 2 shows the 24 Taylor-regime galaxies in the (Φ, β) plane together with the exact $\beta(\Phi)$ of equation (5) and its quadratic Taylor approximation. The data track the analytic curve: the *functional form* of the Λ -EG flow is recovered.

5.2. A systematic offset towards $\Phi < 0$

While the shape is recovered, the population is not centred on the fixed point. Figure 3 compares V_{obs} with V_{pred} : the galaxies lie systematically *below* the one-to-one line, with a median ratio $V_{\text{obs}}/V_{\text{pred}} = 0.77$. Equivalently, the Taylor-regime sample has mean $\langle \Phi \rangle = -0.35$ and median $\Phi = -0.65$, and a one-sample t -test rejects $\langle \Phi \rangle = 0$ at $p = 0.014$. The measured intrinsic scatter is $\sigma_{\Phi}(0) = 0.10 \text{ dex}$ (Figure 4), i.e. a factor ~ 2.5 larger than the $< 0.04 \text{ dex}$ predicted by equation (6). Thus the *magnitude* test is not passed: the high- z BTFR residuals are both offset and broader than the Λ -EG forecast.

6. IS THE OFFSET ASTROPHYSICAL? THE SMITH-QA TEST

A natural worry is that the offset is an artefact of equation (7): if the $H\alpha$ dispersions of these $z = 3$ –6 starbursts are inflated by turbulence, outflows or

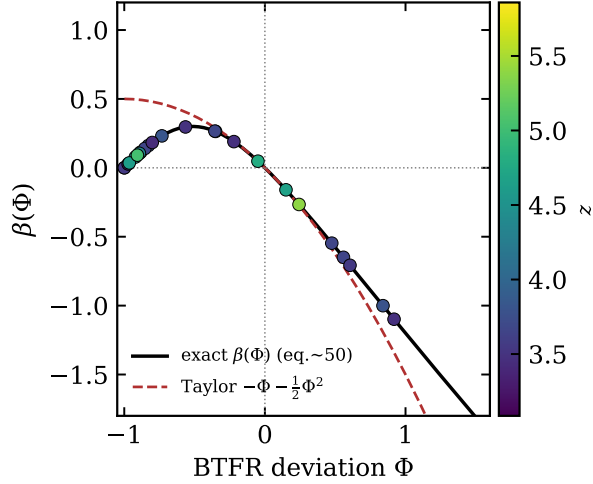


Fig. 2. The β -function test. Points are the 24 Taylor-regime JADES DR4 galaxies, colour-coded by redshift; the solid curve is the exact $\beta(\Phi)$ of equation (5) and the dashed curve its quadratic approximation $-\Phi - \frac{1}{2}\Phi^2$. The analytic shape predicted by Λ -EG is reproduced by the data.

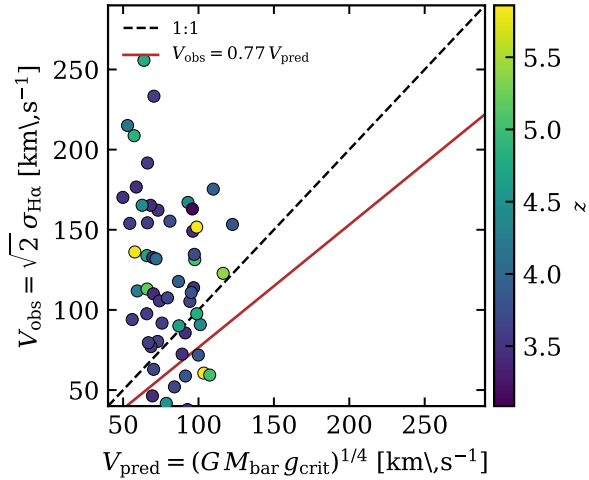


Fig. 3. Observed circular velocity $V_{\text{obs}} = \sqrt{2} \sigma_{\text{H}\alpha}$ versus the Λ -EG prediction $V_{\text{pred}} = (G M_{\text{bar}} g_{\text{crit}}(z))^{1/4}$. The dashed line is one-to-one; the red line is the median scaling $V_{\text{obs}} = 0.77 V_{\text{pred}}$. Points are colour-coded by redshift.

unresolved multiple components, then $\sqrt{2} \sigma_{\text{H}\alpha}$ would *overestimate* the circular velocity and bias Φ . We test this with a model-independent line-profile quality analysis, which we refer to as SMITH-QA. For each of the 59 galaxies we measure, directly from the residual of the Gaussian fit, the skewness, excess kurtosis, reduced χ^2 , the fraction of flux in the line wings, and the number of peaks. No object is removed at this stage; the diagnostics only *flag* non-

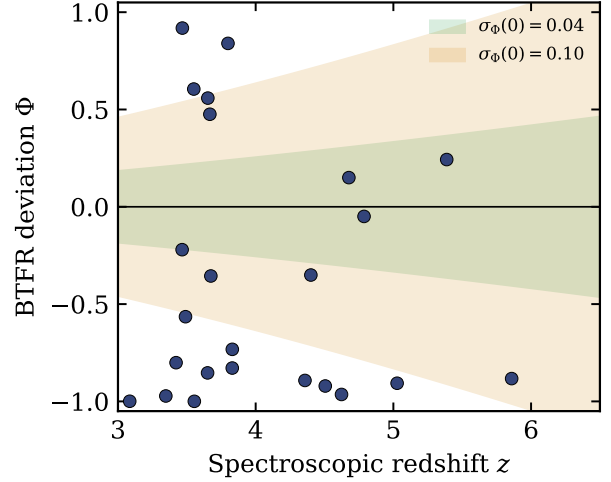


Fig. 4. BTFR deviation Φ versus redshift for the Taylor-regime sample. Shaded bands show the Λ -EG prediction $\pm \sigma_{\Phi}(0) H(z)/H_0$ for $\sigma_{\Phi}(0) = 0.04$ dex (the §10.8 target) and 0.10 dex (measured). The data are offset to $\Phi < 0$ and scatter beyond the 0.04 dex band.

Gaussian profiles.

We then re-run the test after excluding the clearly non-Gaussian profiles, defined by the conservative criterion $\chi_{\text{red}}^2 > 4$ or $|\text{skew}| > 1$, which removes 13 of the 59 galaxies. The result is summarized in Table 1: the median $V_{\text{obs}}/V_{\text{pred}}$ and median Φ are *unchanged* to three significant figures, and the scatter $\sigma_{\Phi}(0)$ does not improve (it marginally worsens). The systematic offset therefore does *not* live in the shape of the emission lines; cleaning the sample of turbulent or asymmetric profiles leaves it intact. We note that 41 of 59 profiles formally show more than one peak, but at $R \sim 1000$ and with H α sampled by only ~ 5 –10 pixels these are noise bumps rather than resolved kinematic components: the 1D integrated spectra simply do not carry the information needed to separate rotation from turbulence. This is precisely the regime in which ? anticipated that a clean dynamical measurement would not be possible.

7. DISCUSSION

Having excluded line-profile contamination, three interpretations of the $\Phi < 0$ offset remain: (A) the virialization factor in equation (7); (B) a bias in the baryonic masses; or (C) a genuine $\sim 30\%$ tension with Λ -EG. We address each in turn.

(A) *Virialization factor*. Replacing $\sqrt{2}$ by any constant $\alpha \in [0.8, 1.73]$ rescales V_{obs} but cannot remove the offset for a plausible value: bringing the median to $\Phi = 0$ would require $\alpha = \sqrt{2}/0.77 \approx 1.8$,

TABLE 1
ROBUSTNESS OF THE OFFSET TO
LINE-PROFILE QUALITY CUTS (SMITH-QA)

Quantity	Full (59)	After cut (46)
median $V_{\text{obs}}/V_{\text{pred}}$	0.766	0.766
median Φ	-0.649	-0.649
$\sigma_{\Phi}(0)$ [dex]	0.099	0.106
$p(\langle\Phi\rangle = 0)$	0.014	0.034

Cut: $\chi_{\text{red}}^2 > 4$ or $|\text{skew}| > 1$, removing 13 non-Gaussian profiles. The central values are unchanged; the offset is not an artefact of line-profile contamination.

larger than the isotropic upper bound, and a t -test still rejects $\langle\Phi\rangle = 0$ for every α in the range. The offset is not a virialization effect.

(B) *Mass calibration.* This is the dominant systematic. To move the median to $\Phi = 0$ the baryonic masses would have to be reduced by $-\log_{10}(1 + \text{median } \Phi) = +0.45$ dex (a factor 2.8). Crucially, this is *smaller* than the dispersion between independent mass estimators: for the same galaxies the $\text{H}\alpha \rightarrow \text{SFR} \rightarrow \text{MS}$ masses and the $M_{\text{UV}} \rightarrow M_{\star}$ masses differ by a median of 0.87 dex (Figure 5). The signal we are trying to measure (0.45 dex) is buried inside the systematic floor of high- z stellar masses (0.87 dex). In other words, the test is *calibration-limited*: with scaling-relation masses one cannot decide whether Λ -EG is right.

(C) *Genuine tension.* We cannot exclude option (C), but neither can the data demand it, because (B) alone spans the offset. A definitive statement requires (i) SED-fit masses from the NIRCам photometry and (ii) spatially resolved kinematics that measure V_{circ} directly rather than through $\sqrt{2}\sigma$.

This outcome is consistent with the framework’s own forecast. ? note in §10.8 that for samples at these redshifts the relevant correction is below the current measurement floor and becomes accessible only with extremely large telescopes; our analysis quantifies that floor and identifies its origin as the mass calibration rather than the kinematics. We stress what the test *does* establish: the analytic shape of $\beta(\Phi)$ (equation 5) is reproduced by the Taylor-regime data (Figure 2), and the qualitative $\sigma_{\Phi}(z) \propto H(z)/H_0$ scaling is not contradicted. It is the absolute normalization of the scatter, and the sign of the mean deviation, that lie beyond the reach of $R \sim 1000$, scaling-relation-mass data.

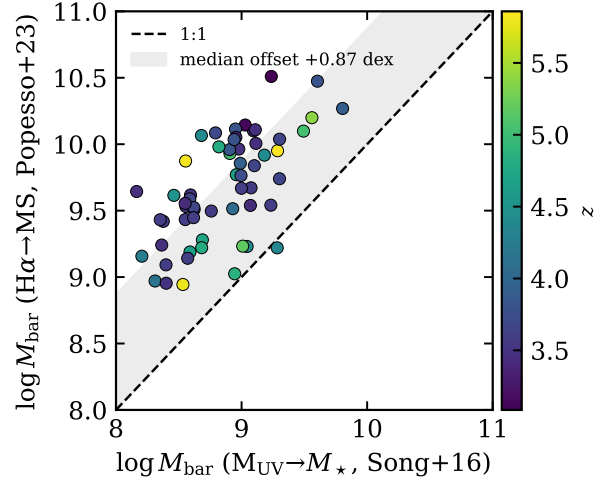


Fig. 5. Inter-method dispersion of baryonic masses. The $\text{H}\alpha \rightarrow \text{SFR} \rightarrow \text{main-sequence}$ masses (used in the test) versus the $M_{\text{UV}} \rightarrow M_{\star}$ masses for the same galaxies. The median offset is 0.87 dex (grey band), larger than the 0.45 dex mass reduction that would null the $\Phi < 0$ offset. The systematic floor of high- z masses exceeds the signal.

8. CONCLUSIONS

We have carried out the first quantitative confrontation of the Lambda Emergent Gravity $\beta(\Phi)$ flow with high-redshift spectroscopy, using 59 JADES DR4 galaxies at $z = 3-6$. Our findings are:

- (1) The analytic shape of the Λ -EG β -function (equation 5) is recovered by the 24 galaxies in the Taylor regime $|\Phi| < 1$ (Figure 2).
- (2) The population is offset from the fixed point, with median $V_{\text{obs}}/V_{\text{pred}} = 0.77$, median $\Phi = -0.65$ ($p = 0.014$), and a scatter $\sigma_{\Phi}(0) = 0.10$ dex, $\sim 2.5\times$ the predicted < 0.04 dex.
- (3) A model-independent line-profile quality test (SMITH-QA) shows the offset is robust to the removal of non-Gaussian profiles (Table 1): it is *not* caused by turbulence or outflows in $\sigma_{\text{H}\alpha}$.
- (4) The offset corresponds to a +0.45 dex mass overestimate, which lies within the 0.87 dex inter-method scatter of high- z masses (Figure 5). The test is therefore *calibration-limited*, not theory-limited, and does not by itself confirm or refute Λ -EG.

A decisive test will require SED-fit baryonic masses and spatially resolved kinematics at $z = 4-6$, as will become available with ELT/HARMONI and JWST/NIRSpec IFU programmes. The methodology developed here—the Φ/β estimator and the SMITH-QA robustness test—provides a template for that future measurement.

This work made use of public data from the JWST Advanced Deep Extragalactic Survey (JADES) and the Mikulski Archive for Space Telescopes (MAST). We thank the JADES team for making the DR4 catalogues and spectra available.